

4 (a) (resultant force = 0) (equilibrium)

therefore: weight – upthrust = force from thin wire (allow tension in wire)

or

$$5.3 \text{ (N)} - \text{upthrust} = 4.8 \text{ (N)}$$

B1 [1]

(b) difference in weight = upthrust or upthrust = 0.5 (N)

$$0.5 = \rho ghA \text{ or } m = 0.5/9.81 \text{ and } V = 5.0 \times 13 \times 10^{-6} \text{ (m}^3\text{)}$$

C1

$$\rho = 0.5 / (9.81 \times 5.0 \times 13 \times 10^{-6})$$

C1

$$= 780 \text{ (784) kg m}^{-3}$$

A1 [3]